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This manual implements Air Force Policy Directive (AFPD) 11-4, *Aviation Service*; AFPD, 11-2, *Aircrew Operations*; AFIC Air Standard ASM 3003 Ed 1v3, *Aviation Medicine/Physiology Training of Aircrew*; and AFIC Information Publication ASM 3009 Ed 1 v3, *Spatial Disorientation*. This manual is consistent with the North Atlantic Treaty Organization (NATO) Standardization Agreement (STANAG) 3114, *Aeromedical Training of Flight Personnel*; NATO STANAG 3474, *Temporary Flying Restrictions Due to Exogenous Factors Affecting Aircrew Efficiency* (**https://nso.nato.int/nso/nsdd/ListPromulg.html**). This publication applies to all units assigned to or gained by major commands (MAJCOM) and Headquarters (HQ) United States Air Force (USAF) direct reporting units (DRU) and Civil Air Patrol (CAP). This applies to commanders, operations supervisors, aircrew, high altitude parachutists, mission essential ground personnel, and passengers assigned or attached to all flying activities of these MAJCOMs and DRUs. This manual applies to all civilian employees and uniformed members of the Regular Air

Force, Air Force Reserve (AFR) and Air National Guard (ANG). This Instruction requires the collection and or maintenance of information protected by the Privacy Act of 1974 (Title 5, United States Code Section 552a) and authorized by Department of Defense Instruction (DoDI) 5400.11, DoD Privacy and Civil Liberties Programs, and DoDI 7730.67, Aviation Incentive Pays and Bonus Program. The applicable System of Records Notices F044 AF SG H, Air Force Aerospace Physiology Training Programs, and F011 AF XO A, Aviation Resource Management System (ARMS) are available: **https://dpcld.defense.gov/privacy/SORNS.aspx**. Ensure all records created as a result of processes prescribed in this publication are maintained in accordance with Air Force Manual (AFMAN) 33-322, Records Management and Information Governance Program, and disposed of in accordance with the Air Force Records Disposition Schedule located in the Air Force Records Information Management System (AFRIMS). Refer recommended changes and questions about this publication to the office of primary responsibility (OPR) using the AF Form 847, *Recommendation for Change of Publication*; route AF Forms 847 from the field through the appropriate functional chain of command. MAJCOMs, and DRUs may supplement this manual as described in **paragraph 1.1.4** The authorities to waive wing/unit level requirements in this publication are identified with a Tier ("T-0, T-1, T-2, T-3") number following the compliance statement. See Air Force Instruction (AFI) 33-360, *Publications and Forms Management*, for a description of the authorities associated with the Tier numbers. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval authority. For non-tiered compliance items, 19 AF Operations Division (19 AF/DO) is the waiver authority for the provisions of this manual. The use of the name or mark of any specific manufacturer, commercial product, commodity or service in this publication does not imply endorsement by the Air Force. Compliance with **Attachment 2** in this publication is mandatory.

(PACAF) AFMAN 11-403, *Aerospace Physiological Training Program,* is supplemented as follows: This supplement applies to all Regular Air Force PACAF aircrew participating in aerospace physiology training in accordance with (IAW) AFMAN 11-403. It also applies only to those Air National Guard (ANG) and Air Force Reserve Command (AFRC) members trained at a PACAF aerospace physiology training unit (APTU). This publication may be supplemented, but all direct supplements must be routed to the office of primary responsibility (OPR) of this publication for coordination prior to certification and approval. Refer recommended changes and questions about this publication to the OPR using the AF Form 847, *Recommendation for Change of Publication*; route AF Forms 847 from the field through the appropriate functional chain of command. The authorities to waive wing/unit level requirements in this publication are identified with a tier number ("T-0, T-1, T-2, T-3") following the compliance statement. See Department of the Air Force Instruction (DAFI) 33-360, *Publications and Forms Management*, for a description of the authorities associated with the Tier numbers. Submit requests for waivers through the chain of command to the appropriate tier waiver approval authority, utilizing guidance identified in DAFI 33-360, or alternately to the publication OPR for non-tiered compliance items. Ensure all records created as a result of processes prescribed in this publication are maintained IAW Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program*, and disposed of in accordance with the Air Force Records Information Management System (AFRIMS) Records Disposition Schedule (RDS). The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force.

SUMMARY OF CHANGES

This document has been substantially revised and needs to be completely reviewed. This revision changes the publication from an AFI to an AFMAN. This manual is written to outline requirements for training, whereas mandatory processes, courseware, and procedures to execute training will be addressed separately in guidance, syllabi, and Technical Orders (T.O.).

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Chapter 1

INTRODUCTION

1.1. Purpose.

1.1.1. This publication prescribes the framework for unit aerospace physiology training programs and procedures, and describes execution of the physiology program specific to aircrew training.

1.1.2. This publication addresses the mission of each unit aerospace physiology training function, and identifies the key processes, tasks and procedures required of the unit in order to accomplish its mission. MAJCOMs will determine the applicability of these publications for mission sets not otherwise specifically addressed by published guidance.

1.1.3. Physiology members includes the total complement of aerospace physiologists and aerospace physiology technicians.

1.1.4. MAJCOMs, DRUs, and FOAs will coordinate their supplement if any to this manual with the Air Education and Training Command (AETC) 19th Air Force Aircrew Performance Branch (19 AF/DOA) before publication and forward one copy to 19 AF/DOA after publication.

1.2. Standard terminology defined.

1.2.1. "Will" and "Shall" indicate a mandatory requirement.

1.2.2. "Should" is normally used to indicate a preferred, but not mandatory, method of accomplishment.

1.2.3. "May" indicates an acceptable or suggested means of accomplishment.

1.2.4. "Note" indicates operating procedures, techniques, etc., which are considered essential to emphasize.

1.2.5. In this manual, aerospace physiology training unit (APTU) refers to either an aerospace physiology training flight (APTF) or aerospace physiology training team (APTT).

1.3. Waiver Authority. Unless otherwise specified, 19 AF/DO is the waiver authority for the provisions of this manual.

1.3.1. Policy and procedures are enacted to provide quality and consistency in training and evaluation whether at an undergraduate or graduate level. Occasionally, unique circumstances may warrant special consideration and possible waiver of policy provisions. Waivers that change the intent of the guidance outlined in this manual are not authorized without 19 AF/DO approval. (T-2).

1.3.2. Request waivers through applicable Operations Group (OG) channels to MAJCOM aerospace physiology program manager (PM), (or equivalent). As applicable, MAJCOM Aerospace Physiology PMs will forward requests to 19 AF/DOA.

Chapter 2

ROLES AND RESPONSIBILITIES

2.1. Headquarters Air Force (HAF).

2.1.1. Deputy Chief of Staff, Operations, Director of Training and Readiness (AF/A3T).

2.1.1.1. Provides guidance and policy for aircrew training in accordance with AFMAN 11-202 vol.1., Aircrew Training.

2.1.1.2. Provides fiscal advocacy to AF Personnel and Training Panel for Planning, Programming and Budgeting requirements to support the Aerospace Physiology (AP) training program.

2.1.1.3. Resolves conflicts in MAJCOM guidance.

2.1.1.4. Designates an AP Aircrew Training PM.

2.1.1.5. Coordinates T-1 compliance waivers with Lead Command to ensure awareness of issues faced by other MAJCOMs in order to make informed programmatic decisions.

2.1.2. The Air Force Surgeon General (AF/SG).

2.1.2.1. Provides qualified personnel in specialty 43AX and 4M0X1 Air Force Specialty Codes (AFSC) to conduct AP training and to operate AP training devices.

2.1.2.2. Ensures 43A and 4M0 recruitment and accessions meet AF/A3T AP Training manpower requirements. Consults with Air Reserve Component (ARC) on reserve accession qualification and training.

2.1.2.3. Coordinates with AF/A3T on any 43A or 4M0 manpower changes exceeding 5% as this affects the AP training program.

2.1.2.4. Provides aeromedical support for the AP training program and provides medical treatment for injuries incurred at locations where training occurs.

2.1.2.5. Provides human performance consultation to AF/A3T and AETC for the AP training program.

2.1.3. The Air Force Chief of Safety (AF/SE). AF/SE, through the Air Force Safety Center (AFSEC), reviews, sanitizes and provides annual mishap trend data, upon request by the MAJCOM PM or MAJCOM Functional Manager (MFM). The AFSEC Office of the Judge Advocate (AFSEC/JA) reviews requests for safety animations and other mishap mitigation material via the Air Force Safety Automated System.

2.2. Headquarters Air Education and Training Command. AETC, as the Lead Command for the AP training program, sets policy and guidance in accordance with AFPD 10-9, Lead Command Designation and Responsibilities for Weapon Systems, and manages the AP training systems in accordance with AFI 16-1007, Management of Air Force Operational Training Systems. AETC has delegated this responsibility to 19 AF.

2.3. 19th Air Force.

2.3.1. Programs and budgets for system-wide unique equipment acquisition, upgrades, and modifications to support the AP training device portfolio. The portfolio includes: altitude

chambers; Reduced Oxygen Breathing Devices (ROBDs); hypoxia familiarization trainers (HFTs); spatial disorientation trainers; and Barany chairs.

2.3.2. Coordinates with the System Program Office (SPO) per AFPD 10-9.

2.3.3. Through Air Force International Affairs (SAF/IA), Air Force Security Assistance Training (AFSAT), and in coordination with AF/A3T, establishes criteria, conducts evaluations and formally recognizes foreign country aircrew AP training programs as meeting USAF requirements.

2.3.4. Coordinates with AF/A3T to recognize and accept non-USAF AP Training for members of other US military services and DoD agencies who fly on USAF aircraft.

2.3.5. Compiles requirements and ensures consistency across the MAJCOMs

2.3.6. As Lead Command, acts in coordination with all affected MAJCOMs. Unresolved differences between MAJCOMs will be elevated to the 19 AF/DO for final resolution.

2.3.7. Designates 19 AF/DOA as the OPR for this manual and execution of the AP Training Program.

2.4. 19th Air Force, Aircrew Performance Branch (19 AF/DOA).

2.4.1. Reviews MAJCOM supplements to this AFMAN to ensure MAJCOMs comply with guidance in this manual.

2.4.2. Maintains liaison with Air Staff organizations, and MAJCOMs to ensure compliance by all physiology personnel.

2.4.3. Coordinates with Air Staff organization and MAJCOM Standardization and Evaluation (Stan/Eval) functions to ensure all separate and subordinate guidance complies with basic Air Force policies contained in this manual.

2.4.4. As the designated OPR for AP training program management, courseware and training systems, coordinates with other organizations to incorporate mishap investigation and prevention information into AP courseware.

2.4.5. Chairs an annual Training Planning Team meeting with MAJCOMs and the AP training SPO.

2.4.6. Develops and communicates AP program objectives and works with MAJCOM PMs and MFMs to define measurable key results to ensure that APTUs and APTTs are focusing efforts on important issues and are meeting aircrew needs.

2.5. Major Commands.

2.5.1. MAJCOMs Director of Operations (MAJCOM/A3). In coordination with the 19 AF/DOA, develops and manages applicable MAJCOM supplements to this manual. MAJCOM supplements will not be less restrictive than guidance contained in this AFMAN. Executes the AP training program in accordance with this publication and any applicable MAJCOM supplements. Coordinates within MAJCOM staff for an AP officer to serve as an AP Training PM for the duties in **paragraph 2.6** Exception: ARC and Air Force Space Command (AFSPC) may appoint an A3 representative without AP or rated restriction due to manpower availability.

2.5.2. MAJCOM Chief of Safety (MAJCOM/SE). Annually reviews and provides MAJCOM Mission Design Series (MDS)-specific human factor and physiological event statistics and trends for incorporation into AP courseware updates. Will provide MAJCOM/A3 AP training PM sanitized mishap descriptions upon request for integration into MDS-specific courseware in accordance with AFI 91-204, Safety Investigation and Hazard Reporting.

2.5.3. MAJCOM Surgeon General (MAJCOM/SG) (with exception for the ARC). Ensures appropriate 43A and 4M0 manning is available to execute AP training program requirements in accordance with this publication or any applicable MAJCOM supplement. Programs and provides resources for professional development of 43A and 4M0 personnel in accordance with Air Force Medical Service (AFMS) career path vectors and Career Field Education and Training Plan (CFETP).

2.5.4. MAJCOM Requirements Directorate (A5 or A8). Ensures all MDS or Aircrew Flight Equipment (AFE) new acquisitions, modifications or upgrades are coordinated with 19 AF/DOA for review for relevancy to incorporate into physiological training courseware. This review includes, but should not be limited to, courseware and training systems to ensure concurrency in addressing MDS and AFE-specific physiological, human factors and emergency procedure training requirements. Coordinates training with the SPO at the Air Force Life Cycle Management Center, Simulators Division (AFLCMC/WNS) for programming required to acquire or modify existing physiological training systems to meet unique requirements as a result of new MDS or AFE acquisitions/modifications/upgrades. Programs for development of physiological training courseware specific to new, modified or upgraded MDS or AFE systems during acquisition process.

2.5.5. MAJCOM aerospace physiology PM and Major Command Function Manager (MFM). Primarily responsible for setting policy and establishing processes within their MAJCOM.

2.5.5.1. Conducts an annual review of physiological training courseware to ensure relevancy to aircrew in the MDS. Forwards recommendations for changes to AP Lead Command. 19 AF/DOA will accomplish this on behalf of the ARC.

2.5.5.2. Participates in annual AP training planning team meetings in accordance with AFI 16-1007.

2.5.5.3. Provides updates to 19 AF/DOA during Aircrew Protection Working Group meetings on any aircrew flight equipment/protective ensembles/night vision systems.

2.5.5.4. Participates in realistic training review boards as required.

2.5.5.4.1. **(Added-PACAF)** Provides feedback to MDS Lead MAJCOM during RAP (Ready Aircrew Program) Tasking Message (RTM) reviews as required.

2.5.5.5. Ensures AP device utilization/maintenance and APTU training reports are submitted to 19 AF/DOA.

2.5.5.6. Assesses applicability/effectiveness of operational aspects of training and ensures compliance with aircrew AP training program with applicable AFIs no less than once every 2 years. Upon request, supports MAJCOM to perform inspections of APTUs.

2.5.5.7. Coordinates supplements to this manual with 19 AF/DOA and provides notification of published MAJCOM supplements.

2.5.5.8. Tracks subordinate unit waiver requests and coordinates with lead commands to allow programmatic adjustments.

2.5.5.9. Convenes conferences and working groups, as necessary, to review and improve command Stan/Eval policies and procedures.

2.5.5.10. Observes execution of unit missions and provide feedback when feasible.

2.5.5.11. Provides functional area compliance inspection guidance to subordinate organizations and MAJCOM staffs. Establishes procedures in MAJCOM supplements.

2.6. Associate Corps Chief for Aerospace Physiology. As the Military Consultant for Aerospace Physiology, works with AF/A3 and AETC to ensure programmatic goals and directives are met. In conjunction with the Air Force Personnel Center (AFPC), ensures that qualified AP personnel are assigned to conduct aircrew and parachutist training.

2.7. Wing Commanders or Equivalent. Provides facilities and resources for executing the AP training program. (T-3).

2.8. Operations Group Commanders (OG/CC) or Equivalent with Assigned Aerospace Physiology Training Units.

2.8.1. Budgets and executes resources for the sustainment of local training devices not managed by the AP SPO (i.e., ROBD and associated gases, Barany chair, lateral drift trainer (LDT)/swing landing trainer (SLT)). Note: sustainment of the altitude chambers (altitude chamber, pumps, and oxygen system) is managed via contracted logistics support by the SPO. The LDT and SLT are considered real property and funded at the local level.

2.8.2. Ensures funding established for AP training is properly distributed to the APTU for execution of training and development of AP personnel.

2.8.3. Ensures documentation of aircrew AP training using existing MAJCOM-approved aircrew training and records management databases.

2.8.4. Provides AP training personnel with appropriate security clearance/access to aircraft flight simulators and training facilities in accordance with AFMAN 11-202 vol.1 and opportunities to fly in assigned aircraft as non-rated aircrew or on non-interference orders per AFI 11-401, Aviation Management.

2.8.5. Ensures aircrew AP Training is MDS and mission-specific and that the training equipment is concurrent with the MDS-specific oxygen equipment and emergency procedures.

2.9. Medical Group Commanders (MDG/CC) with Assigned Aerospace Physiology Training Units.

2.9.1. Ensures OG Commander (CC) or designated representatives are notified of any personnel action or device availability that might affect the execution of AP training requirements. Will ensure appropriate 43A and 4M0 manning is available to ensure AP training program is executed in accordance with this publication or any applicable MAJCOM supplement. (T-3).

2.9.2. Ensures 43A and 4M0 personnel prioritize aircrew/parachutist AP training over participation in medical training/support operations. (T-3).

2.9.3. Programs or provides resources for professional development of 43A and 4M0 personnel in accordance with AFMS career path vectors, CFETP, AFMS Biomedical Science Corps (BSC) flight path and AFI 41-117, Medical Service Officer Education. (T-3).

2.10. Aerospace Physiology Training Unit (APTU) Commanders.

2.10.1. APTUs are designated as regional training units and support aircrew and parachutists from multiple MAJCOMs and wings, as well as ANG and AFR units. APTUs are composed of personnel that support the wing, tenant wing, or other DoD Service MDS-specific aircrew and parachutist physiological training requirements.

2.10.2. Executes the AP training program in accordance with this manual and any applicable MAJCOM supplements. Establishes and maintains a current AP Read File.

2.10.3. Reports all staff and/or student injuries incurred during training in accordance with AFI 91-202 The US Air Force Mishap Prevention Program.

2.10.4. Maintains access to the Air Force Safety Automated System and may provide human performance and human factors analysis on identified hazards in accordance with AFI 91-202. Incorporates AF, MAJCOM, and MDS annual statistics, human factor trends, and physiological mishaps/events into AP training media in coordination with the MAJCOM PM/MFM.

2.10.5. Protects privileged safety information in accordance with AFI 91-204, Chapter 4. (T-0).

2.10.6. Provides AP training schedule as required to supported operational support squadrons (OSS).

2.10.7. Provides fiscal year's AP training schedule to MAJCOM PM & MFM.

2.10.8. Ensures annual validation of hazardous duty orders and monthly chamber flight requirements are met. (T-3).

2.10.9. Reports AP training data to MAJCOM PM & MFM. (T-3).

2.10.10. Develops an altitude chamber/ROBD emergency response plan to address base specific procedures following the removal of a student from a physiology training device due to illness or injury. Units will coordinate and document approval with base Chief of Aerospace Medicine (SGP) annually to review and update the emergency response plan. (T-3).

2.10.11. Supports regional aircrews with AP assets and human performance enhancement training if schedule and manpower availability permit.

2.10.12. Conducts reactor training and emergency procedure training drills. Units with altitude chambers will conduct emergency altitude chamber procedures/drills on a quarterly basis, one of which will involve base first responders. Units with only ROBDs will conduct a minimum of one emergency drill each calendar year which must involve base first responders. Documentation of these activities will be maintained in the unit file plan. (T-3).

Chapter 3

PERSONNEL REQUIRING AEROSPACE PHYSIOLOGY TRAINING

3.1. Personnel Who Require Aerospace Physiology Training.

3.1.1. Aircrew. Aircrew (includes government civil service) will receive training per AFMAN 11-202 vol. 1, and applicable guidance in 11-2 MDS, Volume 1.

3.1.2. High Altitude Parachutists. In accordance with AFI 11-410, Personnel Parachute Operations, AP training will not be used as a screening tool for Free Fall Course selection. (T-1).

3.1.3. Aerospace Physiology Personnel. AFSC 43A and 4M0 providing training in accordance with this manual will maintain physiological training currency. (T-2).

3.1.4. Aircrew Flight Equipment Personnel. AFE personnel in accordance with AFI 11-301 Vol 1, Aircrew Flight Equipment (AFE) Program.

3.1.5. United States Air Force Academy (USAFA) and Reserve Officer Training Corps (ROTC) cadets. Cadets may receive AP training as an instructor or flight trainer position supporting the Airmanship programs.

3.1.6. Contractor Aircrew and Parachutists (including mission crew and flight test engineers). Contractor aircrew and parachutists are required by US Government contract to maintain AP training currency as government furnished training. (T-0). Commanders cannot obligate other units to train their contractors. (T-0). Training must be scheduled by the contract government quality assurance evaluator or government flight representative after verifying the government's responsibility to provide physiological training to the contractor. (T-0). A justification letter must be provided to the APTU as part of the documentation required to receive training. (T-0). Reference **Attachment 3** of this AFMAN for an example of the justification letter.

3.2. Other Personnel Requesting Training.

3.2.1. Deploying Personnel. Personnel may require AP training as part of a deployment tasking. This will be identified by a line remark. Unit Deployment Managers (UDM) may schedule training with the APTU.

3.2.2. Orientation flyers. Orientation flyers identified by host wing operations groups, (including USAFA and ROTC Cadets not identified in paragraph 3.1.5) per AFI 11-401 must meet pre-flight training requirements in AFMAN 11-202, Vol 3, Flight Operations. Additionally, flight surgeons or AP personnel will provide a briefing on physiological symptoms, trapped gas/valsalva maneuver, airsickness prevention, and acceleration forces, if applicable, within 72 hours of flight. If 72 hours are exceeded before flight, students will re-accomplish the training. (T-3).

3.2.2.1. Where available and with qualified operators, the ROBD may be used to familiarize students with their physiological symptoms and provide training in oxygen system use as available if students are medically cleared.

3.2.2.2. AP personnel will not issue an AF Form 1274, Physiological Training to orientation flyers. (T-2).

3.2.2.3. **(Added-PACAF)** Orientation flyers include tandem parachute passengers.

3.2.3. Civilian Distinguished Visitors (DVs). All requests for physiological training of DVs must be coordinated with appropriate OG, MAJCOM/A3 and AF/A3T OPRs through the servicing Public Affairs (PA) office in support of approved PA programs (T-3). The local wing commander may authorize training for civilians when required for military orientation flights. Physiological training of DVs should be associated with a pending aircraft flight or other mission-oriented purpose due to the high risk nature of exposure to reduced partial pressures and/or barometric pressure changes during the training course.

3.2.4. Civil Air Patrol (CAP). CAP members are eligible to receive USAF physiological training. AP personnel may conduct training for CAP on a space-available basis. The minimum age for all non-military training students is 18; comply with the state's age requirement when more restrictive. (T-3).

3.2.5. All Other Requests For Training. Occasionally, specific missions may require members to receive AP training. The member's squadron commander (or equivalent) will assess the benefit of training against the risk and determine if training is required. These requests will be sent to the APTU squadron commander (T-3).

3.3. Scheduling Aerospace Physiology Training. Aircrew and parachutists needing refresher training (**paragraph 4.2.4.4**) should schedule with their closest APTU. Requests for additional training outside of normally scheduled training events must be requested from member's squadron leadership (e.g., CC or Director of Operations) to the training squadron leadership (T-3).

3.3.1. In the event the APTU is unable to support training requests, refer to the Education & Training Course Announcements (ETCA) website: **https://app10-eis.aetc.af.mil/etca/SitePages/ETCA_BCI_CourseDetail_AETC3.aspx?BCIListID=2700** for a listing of APTU locations and contact information.

3.3.2. Aircrew students scheduled for refresher training are required to provide proof of prior AP training. Aircrew students will provide acceptable forms from the Aviation Resource Management System (ARMS), prior AF Form 1274, AF Form 702, or training documentation that shows completion of a course of training which contains Initial Physiological Training as part of the syllabus (e.g., Flight Training Certificate). (T-3). If questions of adequate documentation arise, contact the MAJCOM AP PM or MFM.

3.4. Record Keeping. APTUs will use the 19 AF/DOA-approved student tracker to maintain a record of student training. Units will maintain a record of training for 7 years to support requests for training verification. (T-3).

3.5. Medical Requirements for Aerospace Physiology Training. Individuals must have the appropriate medical clearance in accordance with AFI 48-123, Medical Examinations and Standards to be eligible for physiological training. In the absence of written medical clearance, telephone verification with the student's home medical facility is authorized. Print or type the name of the person who verified the clearance, clearing flight surgeon, and clearance expiration on the flight recorder sheet for that day's training.

3.5.1. US and Foreign Military, Service Academy/ROTC Cadets or Midshipmen, or Government Service Civilians. Copy of DD Form 2992, Medical Recommendation for Flying or Special Operational Duty, indicating that a flying class I, II, or III physical has been

completed using applicable regulatory guidance and containing an expiration date. Cadets shall provide a DD Form 2992 or proof they are medically cleared for hypobaric chamber training within 12 months prior to AP training date. NATO and other foreign military personnel may use the local current base clearance or annual physical DD Form 2992 prepared by home station flight surgeons.

3.5.2. Other. Aircrew who are within 4 months of approved separation, retirement, assigned to a non-flying position, or are placed in an inactive status for administrative or medical reasons, are not required to maintain currency in physiological training.

3.5.3. Personnel without a USAF Flying Class Physical. Government Contractors, Non-DoD Government Civilians and Non-Government Civilians (to include DVs) undergoing USAF physiological training will present a copy of current Federal Aviation Administration (FAA) Medical Certificate flying class I, II or III. The APTU/CC will consult the flight surgeon for final determination of relative health risks of civilians attending AP training.

3.5.4. Refusal of Training. In coordination with the flight surgeon, the APTU/CC may refuse training to any individual that they identify as not meeting the physical health or attitude commensurate with high-risk physiological training.

3.5.5. Physiological Training Personnel. Physiological Training personnel assigned to an APTU are expected to take part in frequent altitude chamber flights as required. The local flight surgeon will evaluate individuals placed in "duty not including flying" (DNIF) status to determine their fitness to perform outside altitude chamber duties. (T-3). If outside altitude chamber duties are authorized, a statement will be annotated in the member's DD form 2992.

3.6. Exclusions from Hypoxia Training.

3.6.1. Exposures after Diving. Physiological symptom recognition training induces hypoxic hypoxia to elicit symptoms. This training entails risk. Personnel must delay altitude chamber and aerial flight exposures for at least 24 hours following dive events (T-2). This includes compressed air or mixed gas diving, surface supplied diving, or hyperbaric chamber exposure. If the dive requires a decompression stop, then 48 hours must elapse prior to altitude chamber exposure. Exception: Special Operations personnel will follow guidelines per the US Navy Diving Manuals. (T-3).

3.6.2. A local flight surgeon will assess a student's medical ability to participate in training if questions arise. (T-3).

3.6.3. Expiring Aerospace Physiology Training during Pregnancy. For aircrew who are medically cleared to fly during pregnancy, altitude chamber and/or ROBD training (see **Table 4.1**) is waived for the duration of the pregnancy. Once medically cleared following pregnancy, aircrew will complete hypoxia exposure prior to first flight (T-3). This will reset expired Aerospace Physiology currency.

3.7. Exceptions to USAF Aerospace Physiology Training.

3.7.1. US Navy Physiology Training. Aircrew who received initial or refresher physiology training for the US Navy, and are current (trained within last 5 years), are not required to complete a full USAF AP training course before their first flight in USAF aircraft.

3.7.1.1. Aircrew who are not current will be required to complete USAF refresher physiology training before their first flight in USAF aircraft.

3.7.1.2. Aircrew members who did not receive altitude chamber exposure for initial training must complete it prior to first flight in USAF aircraft. (T-1).

3.7.2. US Army Physiology Training. Aircrew who received initial or refresher physiology training for the US Army aircrew require additional training only if flying in ejection seat equipped aircraft prior to their first flight in USAF aircraft.

3.7.2.1. Additional training will include the following blocks of instruction:

3.7.2.1.1. Cabin Pressurization

3.7.2.1.2. Crash and Escape

3.7.2.1.3. Acceleration (if flying in high G aircraft)

3.7.2.2. Aircrew who received initial or refresher physiology training from the U.S. Army, and are current (trained within last 5 years) and flying in non-ejection USAF aircraft, are not required to complete a full USAF AP training course before their first flight in a USAF aircraft.

3.7.2.3. Aircrew who are not current will be required to complete USAF refresher physiology training before their first flight in USAF Aircraft.

3.7.2.4. Aircrew members who did not receive altitude chamber exposure for initial training must complete it prior to first flight in USAF aircraft. (T-1).

3.7.3. USAF Exchange Officers. USAF flying personnel on foreign exchange status are not required to complete refresher training prior to returning to USAF aircraft as long as certain conditions are met. They must have received training from countries with recognized USAF physiology training and they must be current on their physiological training. Training documentation must be in English, legible, and certified by a training official from the foreign country. AP personnel will generate an AF Form 1274 to document AP training currency upon receipt of complete training documentation. Use date of training to establish new currency date.

3.7.4. Foreign Aircrew. The USAF recognizes and determines acceptance of physiology training of other countries if the USAF training objectives are met (see **paragraph 3.8**). Rated foreign flying personnel who have current AP training from a recognized country are not required to complete USAF training before their first flight in USAF aircraft. Training documentation and medical clearance must be in English, legible, and certified by a training official from the foreign country. Training must have occurred on or before the country's recognition expiration date. (T-2).

3.7.4.1. Foreign personnel attending a USAF formal flying training program which has physiology training embedded in the syllabus will complete physiology training as part of the syllabus completion. (T-2).

3.7.4.2. Rated aircrew trained in countries after expired approval or with no approval must complete USAF AP refresher training prior to their first flight in USAF aircraft (T-2). Use International Travel Order (ITO) as primary document requiring an individual to obtain USAF AP training; contact Air Force Security Assistance Training Director of Operations (AFSAT/DO) with courtesy copy to 19 AF/DOA to verify training is required and will be resourced for foreign personnel not on ITOs.

3.7.4.3. 19 AF/DOA will maintain the list of approved foreign AP training programs and update quarterly. Questions regarding foreign physiology training should be directed to the Aerospace Physiology Lead Command.

3.7.4.4. The USAF will unconditionally accept AP training, but not centrifuge training, from the United Kingdom, Australia, Canada, and New Zealand. Foreign aircrew members must have completed physiology training including altitude chamber/ROBD training within the last 5 years. 43A or 4M071 or above will sign an AF Form 1274 after verifying prior training documents and completion of additional training requirements. (T-2).

3.8. Foreign Training Program Review.

3.8.1. Requests for USAF approval from other countries must be initiated through the appropriate country manager at SAF/IA and/or AFSAT/DO, who will coordinate with 19 AF/DOA to review the training program.

3.8.2. Upon request from SAF/IA and/or AFSAT/DO, 19 AF/DOA will identify the best qualified AP personnel to review the requesting country's AP training program.

3.8.3. For initial reviews, two person teams (i.e., consisting of field grade 43A3 and 4M071) will complete in-country approval visits after establishing courseware similarity.

3.8.4. Recertification may be accomplished via courseware review at the discretion of 19 AF/DOA. SAF/IA will prioritize and coordinate with AFSAT to fund for country approval requests.

3.8.5. Except for the United Kingdom, Australia, Canada, and New Zealand, foreign AP training programs must be reviewed every 5 years via this same coordination process.

3.8.6. An in-country visit is recommended at least every second review (10 years).

3.8.7. 19 AF/DOA will provide guidance and standardized evaluation checklists to the reviewer(s) to support this process. The reviewer(s) will provide a written summary of their findings and recommendations to 19 AF/DOA for approval.

3.9. USAF Physiology Support to Other Services. The Air Force may provide aircrew AP training support to US Navy, US Coast Guard, and US Army organizations according to support agreements. Military installations and DoD activities may directly communicate with the APTUs to schedule AP training on a space-available basis. USAF aircrew/parachutist training will take priority. (T-3).

3.9.1. **(Added-PACAF)** Squadron Commanders may (as required) establish MOAs with Sister Service mission partners, tenant units or limited-status device locations requesting regular physiological training support. Such agreements should seek to establish clear and manageable training expectations for all parties, based on the capabilities and mission demands of the local APTU. APTUs will provide copies of all finalized MOAs to the PACAF AP Program Manager. **(T-3)**.

Chapter 4

TYPES OF AP TRAINING

4.1. Aerospace Physiology Training Courses. Aerospace Physiology provides specified training courses depending on the member's aircraft flown or the member's AFSC. Table 4.1 illustrates minimum lessons and training events required to complete AP training.

4.2. Aerospace Physiology Standardized Course Material. Aerospace Physiology personnel will teach physiology training addressed in this manual in accordance with the published syllabus and course material available on the AETC Flying Special Publications website, **https://trss3.randolph.af.mil/Bookstore/home/homePage.aspx**. In the absence of a published syllabus, 19 AF/DOA will provide additional guidance. (T-2).

4.2.1. Initial Physiological training delivered as part of an Undergraduate Flying Training (UFT), flying specialty, or AFSC-granting formal training program.

4.2.1.1. With the exception of cadet training, all initial physiological training will be accomplished via the altitude chamber. (T-1).

4.2.1.2. Cadets may receive specialized initial physiology academics as long as time required and scheduling does not interfere with aircrew AP training. Cadet initial training may use ROBD in place of the altitude chamber.

4.2.1.3. Cadets who complete commissioning and selection to an aircrew training pipeline will re-accomplish initial training. For cadets who have commissioned and are awaiting permanent change of station (PCS), initial training is still valid in support of USAFA Airmanship Programs until departure for first assignment.

4.2.1.4. Sister service parachutists meet their respective physiological training requirements provided by their service. Any sister service parachutist member trained by USAF AP will be trained in accordance with requirements in **Table 4.1** (T-1).

4.2.1.5. Written tests are required of all initial students. Test procedures/requirements will be listed in course syllabus. A 43A or 4M071 or above certifies satisfactory initial training using AF Form 1274, upon satisfactory test completion. (T-3).

4.2.1.6. Physiology training requirements are incorporated into the Undergraduate Flying Training syllabus. Students will complete instructor-based initial physiology training requirements specified by 19 AF/DOA as part of their undergraduate course. This includes Career Enlisted Aviators and Special Warfare Airmen. Physiology syllabus components in these flying training courses must meet Initial Aerospace Physiology syllabus requirements as a minimum to receive an AF Form 1274. (T-1).

4.2.1.7. After completing initial training, AF members will complete appropriate refresher training (**paragraph 4.2.2.4**) to maintain currency. Exception: If a member has already completed initial training and attends a formal training course which requires physiology training as part of the syllabus the OG will determine if the member will repeat initial training as part of the syllabus.

4.2.2. Refresher Training. For rated officer aircrew, career enlisted aviators, non-rated aircrew, and AP personnel, refresher training must be completed no later than 5 years from the

end of the month during which initial AP training was accomplished. Any person who has completed initial physiology training (except for USAFA/ROTC cadet) may requalify in AP training by taking the appropriate refresher course for their current flying duties.

4.2.2.1. Refresher training will be conducted in accordance with this manual and the applicable syllabus as developed by 19 AF/DOA (paragraph 4.2.2.4). Training guidance will undergo periodic review.

4.2.2.2. Refresher training may be accomplished with the altitude chamber or ROBD.

4.2.2.3. A 43AX or a 4M071 or above authorized by the MAJCOM AP PM will certify satisfactory refresher training using AF Form 1274.

4.2.2.4. Types of Refreshers. Refresher course categories are grouped based upon common aircraft characteristics and/or AFSCs. To the greatest extent possible, all refresher training should emphasize aircraft-specific oxygen equipment re-familiarization and emergency procedures.

4.2.2.4.1. Track A: For aircrew who fly in high performance/high G aircraft, T-6, and Type 1 & 2 aircraft defined in AFMAN 11-404, Fighter Aircrew Acceleration Training Program. Aircrew who fly in ejection seat-equipped aircraft such as trainer, bomber/attack, fighter or Command Control Communications Computers Intelligence Surveillance and Reconnaissance (C4ISR), except for aircrew flying U-2.

4.2.2.4.2. Track T: For aircrew who fly in non-high G aircraft. For aircrew who fly in tanker/transport aircraft, and manned C4ISR aircraft that do not fly with an ejection seat. Note: USAF CV-22 aircrew may attend either a Track T or H course.

4.2.2.4.3. Track H: For aircrew who fly in helicopter/rotary wing aircraft.

4.2.2.4.4. Track J: For parachutists who have previously completed high altitude parachutist (HAP) Initial physiological training and are Military Free Fall (MFF) qualified.

4.2.2.4.4.1. **(Added-PACAF)** Sister Service members who completed their service specific initial physiological training via ROBD may attend Track J refresher training assuming all other training criteria are met (i.e. proof of prior training, medical clearance verification). Sister Service members who completed their service specific initial physiological training and are projected to attend Military Free Fall (MFF) are authorized to participate in Track J refreshers.

4.2.2.4.5. Track R: For Remotely Piloted Aircraft (RPA) aircrew. No altitude chamber or ROBD training is required.

4.2.2.4.6. Track E: For E-9 Group Superintendent (or higher) or O-6 Group Commander (or higher) aircrew members or parachutists. Hypoxia practical is optional. If student opts for academics only training, academics may be completed by a live video teleconference.

4.2.2.5. With the exception of some Track E refresher students (see **paragraph 4.2.2.4.6**) and Track R students, all refresher students will complete hypoxia exposure via altitude chamber or ROBD regardless of number of years of flying service. (T-1).

4.2.3. The MAJCOM/A3 may establish more frequent MDS-specific training requirements as appropriate. Reference aircraft-specific 11-series publications and MAJCOM supplements for more information on these additional training requirements.

4.2.3.1. **(Added-PACAF)** MDS-Specific Training. Conduct additional AP training as required by specific 11-2MDS series guidance or applicable directives. MDS-specific training is meant to supplement already specified training requirements and does not replace the 5-year currency requirement for refresher physiology training. For example, physiological event training conducted via ROBD (Task ID PP10) is required every 12 months for F-22 and F-35 aircrew.

4.2.3.2. **(Added-PACAF)** F-22 high altitude Pressure Breathing for Altitude (PBA) training academics only (Task ID PP11) may be accomplished at the same time as F-22 Track A refreshers ACC/A3T is the OPR for these academics, and will consult with ACC AP PM and ACC TRSS/HP to update PBA initial and refresher course material.

4.2.3.3. **(Added-PACAF)** For contingency deployments, MDS-specific AP training (i.e., PP10) is extended to 18 months to ensure currency beyond the length of the deployment. Aircrew unable to meet annual requirements (e.g., ANG/Reserve units without AP assets) may request waiver and/or currency extension with local OG/CC **(T-3)**.

4.2.3.4. **(Added-PACAF)** Practical and academic training for Task ID PP10 provides an annual aircrew refresher for physiological symptoms and hypoxia-like conditions (e.g., physiological events), to include pilot familiarization with aircraft-specific life support and backup/emergency oxygen systems.

4.2.3.4.1. **(Added-PACAF)** Annual refresher training (Task ID PP10) consists of MDS-specific oxygen system information (e.g. BOS/EOS), PE trends, current physiological research and mitigation efforts, followed by an ROBD training profile programmed for a 25,000' hypoxia demonstration. MDS-specific oxygen systems and procedures will be reinforced concurrently with ROBD training profile. No additional night vision demonstration is required unless being accomplished in conjunction with 5-year refresher training.

4.2.3.4.2. **(Added-PACAF)** Aircrew completing the 5-year Refresher Physiological training (Track A) may also receive credit for the annual physiological event training (Task ID PP10) if additional MDS-specific oxygen system objectives are met. In addition to the AF Form 1274 *Physiological Training*, documenting refresher training accomplishment, aircrew will also be provided an AF Form 1522, *ARMS Additional Training Accomplishment Report*, to document completion of PP10 and PP11.

4.2.4. Courseware is restricted from release without approval of 19 AF/DOA.

4.2.5. Air Combat Command (ACC) will provide guidance for Pressure Suit training in support of U-2 operations.

Table 4.1. AP Training Requirements.

	Initial	HAP Initial	RPA Initial
A. Physiological Effects of Altitude	X	X	
B. Performance Threats	X	X	X

C. Physiology Considerations of Aircrew Flight Equipment	X		X			
D. Cabin Pressurization/Depressurization	X					
E. Vision	X		X			
F. Spatial Disorientation	X					X
G. Noise and Vibration	X		X			
H. Attention Management Threats to Situational Awareness (SA)	X		X			X
I. Acceleration	X					
J. Physiology Considerations of Aircraft Egress	X					
K. Altitude Chamber	X		1			
L. ROBD						
M. Barany Chair/Spatial Disorientation (SD) Trainer	X					
N. Unaided Night Vision Trainer	X		X			
Refresher Training Requirements by Tracks						
	A	T	H	J	R	E
O. Physiological Effects of Altitude	X	X	X	X		X
P. Performance Threats	X	X	X	X	X	
Q. Physiology Considerations of Aircrew Flight Equipment	2	2	2	2		
R. Cabin Pressurization/Depressurization						
S. Vision	X	X	X	X	X	
T. Spatial Disorientation	X	X	X		X	X
U. Noise and Vibration	X	X	X	X	X	
V. Attention Management Threats to SA	X	X	X	X	X	X
W. Acceleration	X					X
X. Physiology Considerations of Aircraft Egress						
Y. Altitude Chamber	4	4	4	4		3
Z. ROBD	4	4	4	4		3
AA. Barany Chair/SD Trainer						
AB. Unaided Night Vision Trainer						

Notes:
Note 1: USAF parachutist training requires initial altitude chamber exposure. Parachutists from other branches of service will meet their respective service specific requirements.
Note 2: MAJCOM/A3T identified Special Interest Item/Flight Crew Information File or topic as required.
Note 3: Hypoxia practical is optional.
Note 4: Refresher training may use altitude chamber or ROBD to meet hypoxia training requirements.

4.3. (Added-PACAF) Other Aircrew Training Supported by AP personnel. AP personnel may support the following: Night Vision Device (NVD) Training, Cockpit/Crew Resource Management (CRM), Instrument Refresher Courses, Anti-G Straining Maneuver (AGSM) Heads Up Display (HUD) reviews, familiarization/orientation flight briefings, squadron and wing safety briefings, etc. IAW applicable instructions or manuals governing these programs.

4.3.1. **(Added-PACAF)** Where available, Aerospace Physiologists will participate in Unit's Gravitational load factor (G)-Awareness CT program IAW 11-2MDS guidance. Tactical mission recording reviews include assessment of the AGSM and human factor threats. Human factors include, but are not limited to, pilot breathing patterns and MDS-specific attention management threats. Local commanders will ensure a minimum of one AP team member has SAP clearance to support Unit's Gravitational load factor (G)-Awareness CT program.

Chapter 5

AEROSPACE PHYSIOLOGY TRAINING OPERATIONS

5.1. Qualified Personnel. Trained and qualified aerospace physiology personnel, as well as personnel in upgrade or qualification training under supervision, will provide aerospace physiology training. (T-1). Personnel will be trained and qualified in accordance with Aerospace Physiology published guidance. Until formal guidance is published units will use the 43AX or 4M0X1 CFETP, 19AF/DOA guidance, and MAJCOM guidance to standardize the training and qualification process. (T-2).

5.2. Physiology Training Content. Physiology training consists of a live academic portion and a practical hypoxia exposure via the altitude chamber or ROBD. For altitude chamber crew complement, reference T.O. 43D8-3-2-81 Hypobaric Training Assembly.

5.3. Aerospace Physiology Training Devices. Operate all AP training devices in accordance with this manual, T.O.s, and local unit Operating Instructions.

5.3.1. Local unit instructions will follow Air Force Occupational Safety and Health (AFOSH) standards. (T-1).

5.3.2. With the exception of **paragraph 5.6**, use of the physiology training devices for experimental activities is not authorized without 19 AF/DOA approval. (T-2).

5.4. Special Flight Profiles. MAJCOM AP PM/MFM, 4M0X1 may develop special flight profiles to meet specific training requirements. 19 AF/DOA must approve these profiles before they are used.

5.5. Completion of Device Training. Practical, performance-based training using devices is optimally delivered immediately following academic instruction. Students who do not complete device training within 90 calendar days of academic training must repeat AP training in its entirety. (T-2).

5.6. Medical Evaluation Chamber Flights. Medical evaluation chamber flights require direct supervision of the attending flight surgeon. Unique flight profiles to evaluate individual patients will be determined by the flight surgeon, in consultation with the Aerospace Physiology Officer (APO). Medical evaluation flights will not interfere with operational training. The flights will not expose any person to altitudes at or above FL250 for more than 30 minutes or above FL180 for more than 1 hour due to the increased risk for decompression sickness (DCS). (T-3).

5.6.1. Minimum crew complement requirements and inside observer exposure limits apply to this non-training use of the altitude chamber in accordance with T.O. 43D8-3-2-81. (T-2).

5.6.2. A flight surgeon may accompany a qualified inside observer to directly monitor medical evaluation flights. (T-3).

5.7. Student Disputes. Students who show unprofessional behavior during physiology training will receive a non-completion letter from the AP flight commander and will be referred to their leadership for further action. (T-3).

Chapter 6

ACADEMIC TRAINING

6.1. Aerospace Physiology Training Purpose. The purpose of AP training is to provide aircrew and parachutists the knowledge, skills, and attitudes to optimize performance in challenging air and space operations. AP training systems provide a training environment that replicates high altitude, high-G, egress, and orientation-compromised exposures in which personnel can apply preventive countermeasures and practice emergency procedures. Providing academic lessons followed by hands-on application in a controlled environment, maximizes student learning while minimizing risk exposure. All AP personnel are expected to maintain familiarity with the missions, relevant systems (e.g., oxygen equipment), and modifications (e.g., display upgrades) of USAF major weapon systems represented in initial and refresher AP training.

6.2. Flying Training Courseware.

6.2.1. AP courseware will be posted on the AETC Undergraduate Flying Training Special Publications site for instructor use. All AP instructors shall have access to courseware. Use the Course Change Request (CCR) through the bookstore to submit recommendations for changes. Requests for access to the AETC Flying Training Special Publications site may be made on the site itself at **https://trss3.randolph.af.mil/Bookstore/home/homePage.aspx**.

6.2.2. Information contained in flying training publications is identified as controlled unclassified information (CUI) due to MDS technical and operational content. This creates the requirement for the accessibility and releasability statements found in front of the publications.

6.2.3. Organizations may receive unsolicited requests for syllabi, lesson plans, or other courseware documents. Refer requests for release to 19 AF/DOA and 19 AF/JA for legal review.

6.2.4. Approval from 19 AF/DOA is required for release of all courseware. (T-2).

6.2.4.1. **(Added-PACAF)** Courseware will be reviewed and updated annually to ensure current, relevant and applicable safety and mishap data is included. AP personnel will include local MDS-specific safety data into their courseware (e.g. ASAP, MFOQA, AFSAS mishaps and LOSA reports). A combination of MAAF videos and Proactive Aviation Safety events will be used as case studies to reinforce each block of instruction. Privileged safety information will be handled IAW AFI 91-204.

6.3. Academic Lessons. Personnel providing academic instruction to students must be trained and qualified in accordance with AFSC CFETP, 19 AF/DOA, and MAJCOM guidance. (T-2). Certain subjects require specific AFSCs or skill level in order to teach. See **Table 6.1** for a breakout of academic blocks of instruction and AFSC required to teach. (T-2).

Table 6.1. Teaching Qualification Requirements.

	4M031	4M051	4M071 and above	43AX
X= CFETP qualification required to teach (See Note 1) A= Authorized to teach (See Note 2)				
Physiological Effects of Altitude		A	A	X
Performance Threats		A	A	X
Physiology Considerations of Aircrew Flight Equipment	X	X	X	A
Cabin Pressurization and Decompression	X	X	X	A
Pressure Breathing	X	X	X	X
Vision/Unaided Night Vision		A	A	X
Spatial Disorientation		A	A	X
Noise and Vibration		A	A	X
Attention Management Threats to SA		A	A	X
Acceleration		A	A	X
Physiology Considerations of Aircraft Egress	X	X	X	A
Barany Chair/Spatial Disorientation Trainer	X	X	X	X
Altitude Chamber Lecture	X	X	X	A
ROBD Lecture		A	A	X

Notes:
Note 1. Individuals required to teach specific blocks of instructions will maintain qualification in those blocks IAW AFSC CFETP, 19 AF/DOA, and MAJCOM guidance until specific career field training and qualification guidance is published.
Note 2. Individuals authorized to teach specific blocks of instruction must meet training/qualification requirements in accordance with 19 AF/DOA and MAJCOM guidance, until specific career field training and qualification guidance is published, and have the approval of the APTF/APTT Commander/Chief.

Chapter 7

ALTITUDE CHAMBER OPERATIONS

7.1. Altitude (Hypobaric) Chamber. The altitude (hypobaric) chamber provides a training system which replicates the effects of barometric pressure change on the human body. The effects include exposure to low barometric pressure environment for recognition of personal hypoxia symptoms as well as physical effects of pressure change at various training altitudes.

7.1.1. The altitude chamber will be operated in accordance with TO 43D8-3-2-81 (T-1).

7.1.2. Aerospace Physiology personnel will not perform inspections from the top of the altitude chamber. (T-3). The only component on the top of the altitude chamber that currently requires visual inspection is the rapid decompression solenoid valve; rapid decompression solenoid inspection should be executed from ground level or using a portable ladder.

7.2. Personnel Requirements.

7.2.1. Personnel requirements for operation of the altitude chamber are listed in TO 43D8-3-2-81. Lock operator position is not required for an altitude chamber flight with a single student.

7.2.2. A qualified APO is required to supervise an altitude chamber flight anytime it climbs with a person inside. (T-2).

7.2.3. Personnel filling altitude chamber crew positions must be current in cardiopulmonary resuscitation (CPR) training to include use of the automated external defibrillator (AED). (T-3).

7.2.4. Exposure Limitations.

7.2.4.1. Maximum exposures for Inside Instructors-Observers inside hypobaric chambers:

7.2.4.2. Two flights in a 7-day period to or above 25,000 feet. (T-2).

7.2.4.3. Two rapid decompressions in a 7-day period. Rapid decompression flights may be taken in combination with exposures to or above 25,000 feet. (T-2).

7.2.4.4. At least 72 hours must elapse between exposures to altitudes to or above 18,000 feet. (T-2).

7.2.4.5. At least 23 hours will elapse between exposures to rapid decompressions below 18,000 feet. (T-2).

7.2.4.6. Inside personnel or students will not exceed altitudes at or above FL250 for more than 30 minutes or above FL180 for more than 1 hour.

7.2.4.7. The flight surgeon will be notified and respond to altitude chamber reactions in accordance with locally developed response plans. (T-3).

7.2.4.8. Squadron commanders may approve chamber crew compliment variations if manpower constraints (number of assigned/qualified, DNIF, etc.) prevent the unit from operating within exposure limitations. Commanders will not approve blanket or group variations, but instead authorize on a case by case basis. (T-2).

7.2.4.9. The AP flight commander will notify the MAJCOM PM and MFM when altitude chamber crew complement is reduced. (T-2).

7.3. Fire Safety.

7.3.1. Due to repeated exposure to fire risks, all crewmembers will wear approved flight clothing (flight suit and boots) while performing altitude chamber or ROBD duties. (T-3). Exception: Maternity utility uniforms are acceptable when performing outside crew positions and ROBD duties when necessary.

7.3.2. AP personnel servicing pressurized gas containers will wear nomex gloves, approved flight clothing and eye protection. Exception: Maternity utility uniforms are acceptable when performing outside crew positions and ROBD duties when necessary. Gloves should be clean of dirt or grease stains to maximize the flame resistance properties of the material. (T-3). Closed and appropriately capped pressurized gas containers being moved in banks do not require personal protection equipment (PPE) wear. (T-3).

7.4. Altitude Chamber Flights.

7.4.1. Pre-flight Briefing. Students will receive altitude chamber flight pre-brief and post-briefings prior to and following the altitude chamber flight.

7.4.2. Altitude Chamber Pre-flight Briefing. At minimum, all students will be briefed on the following altitude chamber flight objectives prior to altitude chamber training. (T-2).

7.4.2.1. Accomplishing all steps for complete hypoxia recovery.

7.4.2.2. Exhibit proper techniques to combat mechanical effects of pressure change for ear, sinus and gastrointestinal gas expansion.

7.4.2.3. Demonstrate proper oxygen equipment discipline and in-flight oxygen checks.

7.4.2.4. Identify and use emergency/portable oxygen equipment.

7.4.2.5. Recognize and correct night vision deficiencies resulting from decreased oxygen.

7.4.2.6. Determine physical indication and physiological effects of a rapid decompression, only required for initial students.

7.4.2.7. Ensure students understand when, how, and why to use the “Level Off” or “Knock it Off” hand signal.

7.4.2.8. Ensure students know the steps to perform an effective Valsalva maneuver.

7.4.3. Crew Brief. AP personnel may conduct a pre-flight crew briefing for each altitude chamber flight. Minimum personnel involved should be the crew members assigned to the altitude chamber flight.

7.4.3.1. The crew pre-flight brief may consist of the following:

7.4.3.2. Crew member position identification.

7.4.3.3. Altitude chamber maintenance discrepancies.

7.4.3.4. Type of altitude chamber flight.

7.4.3.5. Flight profile.

7.4.3.6. Number of students.

7.4.3.7. Student issues.

7.4.3.8. Crew members in upgrade training.

7.4.3.9. Crew members receiving a qualification evaluation.

7.4.3.10. Crew member actions during emergency. As a minimum the crew will discuss each member's response to one mechanical emergency and one medical emergency.

7.4.4. Altitude Chamber Reactor Reporting.

7.4.4.1. Because not all altitude chamber reactions meet safety reporting criteria, altitude chamber reactions will be tracked using the grading scale in **Table 7.1** and will include the anatomical location of the reaction (right ear, abdominal pain, etc.). APTUs will review reaction data for trends as appropriate, include this information in instructor training meetings-and annual reports as required. Units will not maintain any health or Privacy Act information on reactions, only information useful in identifying and documenting trends.

7.4.4.2. All confirmed DCS cases resulting from altitude chamber training are managed as an occupational illness. (T-2).

Table 7.1. Altitude Chamber Reactor Grading.

Grade	Description
1	Requires no level-off or change in flight profile. Treatment is limited to verbal inquiry or instruction only. An example of a grade 1 reaction would be instructing someone to Valsalva on descent if member complains of fullness in their ear and Valsalva results in alleviation of the problem.
2	Interrupts a flight profile or levels the altitude chamber. An example of a grade 2 reaction would be a level of the altitude chamber to resolve an ear problem or an increase of up to 2,000 feet (also known as a bounce) of the altitude chamber to relieve ear pain.
3	Requires removal from the altitude chamber and/or placed on 100% oxygen for observation without hospitalization. DCS symptoms are classified as at least a grade 3 reaction. Includes most reactions requiring flight surgeon consult. An example of grade 3 would be removal of a student from the altitude chamber due to gas pains at altitude or administering surface level oxygen for suspected DCS.
4	Requires admission to the hospital or treatment in the hyperbaric chamber. All Grade 4 reactions require flight surgeon consult and must be reported to the local Wing Safety Office.

7.4.5. Post-flight Briefing. At minimum, all students will be briefed on the following topics following altitude chamber training. (T-2).

7.4.5.1. Students should not participate in physical exercise, strenuous activity or extended duty for a period of 12 hours.

7.4.5.2. Students may fly as crewmembers or passengers after an altitude chamber flight to 25,000 feet or below but should remain below a cabin altitude of 15,000 feet.

7.4.5.3. Students should perform periodic Valsalva maneuvers throughout the day/evening to prevent delayed ear blocks.

7.4.5.4. Students should avoid alcohol consumption for 12 hours after a hypobaric exposure. Alcohol can mask DCS symptoms, and dehydration can increase DCS risk.

7.4.5.5. Students should monitor their fatigue level after a hypobaric exposure.

7.4.5.6. Provide students contact information and procedures to report suspected DCS while in the local area, outside the local area, and after hours.

7.4.6. The altitude chamber crew will conduct a post-flight briefing. During the post-flight brief the crew will discuss the following:

7.4.6.1. Deviations from briefed profile.

7.4.6.2. Maintenance discrepancies. Record discrepancies on appropriate maintenance documentation (e.g., AFTO 244, Industrial/Support Equipment Record).

7.4.6.3. Equipment discrepancies.

7.4.6.4. Improvement areas. Lessons learned may be submitted to 19 AF/DOA for use as a training resource for other APTUs.

Chapter 8

REDUCED OXYGEN BREATHING DEVICE (ROBD) OPERATIONS

8.1. Reduced Oxygen Breathing Device Training Systems. Within the context of USAF physiology training and this manual the term ROBD is used to describe a training system that provides hypoxia recognition and emergency procedures training using normobaric-reduced oxygen gas mixtures. It is not intended to define a brand name.

8.2. Refresher Training Use. For those personnel requiring refresher training, it is an acceptable training system for physiological symptom training. ROBD training delivers the reduced oxygen gas mixture to the student and is often referred to as 'Mask-On' hypoxia training. ROBD training allows some students to wear their primary oxygen equipment.

8.3. Reduced Oxygen Breathing Device Set-Up. The ROBD is to be used in conjunction with a Hypoxia Familiarization Trainer Crew Cockpit (HFT-C) or Mission Crew Trainer (HFT-M). Locations where aircraft flight simulators can accommodate ROBD may use MDS-specific oxygen systems and emergency procedures to improve training realism.

8.4. Hypoxia Familiarization Trainer (HFT). The HFT provides a flight task simulation for use in conjunction with an ROBD during which the aircrew can undergo hypoxia recognition and oxygen system emergency procedures training. Two configurations are available, the HFT-C or cockpit trainer and the HFT-M for mission crew. Training scenarios should enable the aircrew to practice use of their primary oxygen system, or any emergency or alternate oxygen systems to the maximum extent possible. Helmet and masks are not provided by APTT locations, students must bring their own AFE gear. APTT personnel may coordinate local support agreements for AFE gear with host wing AFE.

8.4.1. The HFT-C is a generic flight simulator which is configurable to represent the oxygen regulator position of the student's aircraft to provide more realistic training.

8.4.2. The HFT-M is designed for non-pilot crewmembers. It uses cognitive performance software to demonstrate reduction of performance during the hypoxia exposure. The HFT-C can be used by non-pilot crewmembers to complete training.

8.5. Reduced Oxygen Breathing Device Training Plan. ROBD training will be accomplished using the host wing OG-approved training plan. Students will receive ROBD pre-brief and post-briefings prior to and following the training system event. (T-2).

8.6. Reduced Oxygen Breathing Device Guidance. Personnel will operate the ROBD in accordance with applicable manufacturer operating manuals, or guidance provided by 19 AF/DOA.

8.7. Personnel Requirements.

8.7.1. Minimum Personnel Requirements for ROBD: AP personnel, AFSC 43A1 or 4M051 or above, may instruct and operate the ROBD for aircrew training. If several ROBDs (two or more devices) are being used in the same location, each device will have a qualified and dedicated Instructor/Operator (I/O). All I/Os will be trained and qualified in accordance with AFSC CFETP, 19 AF/DOA, and MAJCOM guidance until specific career field training and qualification guidance is published.

8.7.2. Personnel filling ROBD I/O positions must be current in CPR training to include use of the AED. (T-3).

8.8. Reduced Oxygen Breathing Device Pre-flight Briefing. At minimum, all students will be briefed on the following topics before completing ROBD training.

8.8.1. Provide students with an ROBD orientation.

8.8.2. Describe the following training objectives:

8.8.2.1. Experience hypoxia while in a simulated flight/mission environment.

8.8.2.2. Perform the appropriate hypoxia emergency procedures without assistance.

8.8.2.3. Demonstrate correct positive pressure breathing technique during emergency oxygen activation, if supplemental pressure is used.

8.8.2.4. Demonstrate pre-flight, in-flight, emergency, and post flight equipment configuration checks.

8.8.2.5. Recognize and correct night vision deficiencies resulting from decreased oxygen.

8.8.2.6. Recognize the negative effect of hypoxia on effectively completing mission-related duties.

8.8.2.7. Review the ROBD profile and simulator, HFT or part task trainer scenario as appropriate.

8.9. Reduced Oxygen Breathing Device Post-flight Briefing. At minimum, all students will be briefed on the following topics following ROBD training.

8.9.1. Advise students to contact a flight surgeon or emergency treatment facility in the event unusual symptoms occur after the completion of ROBD training.

8.9.2. Advise students that there are no aircrew, jump, or passenger flight restrictions placed on personnel following ROBD training.

8.9.3. Students should monitor their fatigue level after ROBD training.

Chapter 9

OTHER TRAINING DEVICES

9.1. Other Aerospace Physiology Training Devices. Some AP training devices are used solely within a specific MAJCOM (e.g., pressure breathing training for F-22). Guidance for these devices should be provided via MAJCOM supplement to this publication. Configuration control will be maintained by the respective MAJCOM AP PM/MFM, 4M0X1; however devices used by more than one MAJCOM will be maintained by 19 AF/DOA. If devices are purchased for use by other MAJCOMs, then device guidance will be issued via a change to this publication.

9.2. Barany Chair. The Barany Chair is used as introductory spatial disorientation demonstrator and for rotational training as part of the airsickness management program. Operate the Barany Chair in accordance with operating instructions published by 19 AF/DOA and MAJCOM supplementing guidance.

9.3. Unaided Night Vision Trainer. The unaided night vision trainer is used during initial physiological training to demonstrate unaided night vision limitations. Operate the Unaided Night Vision Trainer in accordance with operating instructions published by 19 AF/DOA and MAJCOM supplementing guidance.

9.4. Spatial Disorientation (SD) Trainer. The spatial disorientation trainer, or GYRO IPT II, is used during initial physiological training for the undergraduate pilot training and pilot instructor training pipelines. Operate the spatial disorientation trainer in accordance with operating instructions published by 19 AF/DOA.

Chapter 10

DEVICE MANAGEMENT

10.1. General. 19 AF/DOA coordinates with the AP Training Devices Program Office to ensure configuration management for managed AP training devices per AFPD 10-9. 19 AF/DOA will develop a strategic plan for training devices.

10.1.1. APTU personnel are responsible for local management of AP training devices and ensure all technical manuals/orders, associated checklists, and procedures are properly followed for storage, operation and maintenance.

10.1.1.1. **(Added-PACAF)** AP training devices will not be shipped or transported for use at locations other than where assigned without prior coordination with PACAF AP PM and AP Lead Command office.

10.1.2. APTUs requesting new devices in addition to the strategic plan may request them via AF Form 1067 Modification Proposal through the MAJCOM AP PM/MFM to 19 AF/DOA. Requests will document maintenance and consumables funding, manning support to the device, and if requestor is asking 19 AF/DOA to provide contracting liaison.

10.2. Device Status Reporting. Per AFI 16-1007, 19 AF/DOA is responsible for managing training device utilization. APTUs with ROBD capability will report local device utilization and maintenance information for the preceding month to 19 AF/DOA no later than the 10th day of each month. Complete the following forms using the guidance provided by 19 AF/DOA.

10.2.1. 19 AF/DOA will compile and provide annual fiscal year usage data as needed to AF/A3T and the SPO in accordance with this manual.

10.2.2. APTUs will use an AF Form 4026, Aircrew Training Devices Utilization, to report monthly and annual use of altitude chambers, pumps, compressor, SD trainers, ROBD and HFT. APTUs may submit one completed AF Form 4026 to report all devices they operate.

10.2.3. APTUs will use AFTO Form 244 to report maintenance discrepancies and resolution for use of the altitude chamber and vacuum pumps. When all lines of the AFTO Form 244 are completed send copy to 19 AF/DOA through the AETC SharePoint™.

10.2.4. APTUs will use AFTO Form 95 Significant Historical Data to maintain historical record of the altitude chamber and vacuum pumps. When all lines of the AFTO Form 95 are completed send copy to 19 AF/DOA through the AETC SharePoint™.

10.2.5. Original AFTO Forms 244 and 95 will remain with the device in the event of relocation or are turned in when a device is removed from service.

10.2.6. Use the AFTO Form 334, Helmet and Oxygen Mask/Connector Inspection Data, to document maintenance actions on appropriate aircrew flight equipment. This document will be maintained locally and will not be submitted to 19 AF/DOA.

10.2.7. Units providing AP training will notify their respective wing and MAJCOM AP POCs, and 19 AF/DOA when AP training devices are non-operational due to maintenance or otherwise not an available training option for wing students within 24 hours of non-operational status. (T-2).

10.2.8. **(Added-PACAF)** Device Management for ROBD operations without locally assigned AP Personnel. Local squadrons or groups without locally assigned AP personnel assume responsibility for meeting all requirements associated with AP training equipment sustainment and safe operations IAW PACAF AP PM coordination. Squadrons with vacant AP positions may contact the PACAF AP PM for guidance in device sustainment and/or proper storage.

10.3. Device Maintenance. Maintenance not detailed within the appropriate technical manuals will not be accomplished unless coordinated with 19 AF/DOA. Note: All personnel will comply with T.O.s and manuals in accomplishing maintenance (i.e., daily, periodic, special inspections etc.). (T-1).

10.3.1. Altitude chamber discrepancies are to be handled and documented by AP personnel to the maximum extent possible using proper T.O.s and the appropriate forms, respectively. (T-2).

10.3.2. If an altitude chamber problem is encountered that is not addressed by a T.O. or is beyond the scope of local expertise, the local AP contracting officer's representative (COR) will contact the altitude chamber contract maintenance provider. Contact MAJCOM AP PM/MFM, 4M0X1 for maintenance guidance on other devices.

10.3.2.1. Contractors may request photographs or additional information on local issue to provide help desk support. This information should be provided as quickly as possible to ensure rapid support. APTUs are not expected to provide information technology (IT) support to contractors during site visits. Units will provide base access on behalf of the contract team.

10.3.2.2. If the contractor is required to accomplish maintenance procedures, the team will not depart the site until a government acceptance check is completed. The government acceptance check is required to verify that the altitude chamber is fully operational after all corrective actions are complete. The COR will ensure proper documentation is completed prior to the departure of the maintenance team. As a minimum, use the daily inspection checklist in the T.O. to verify altitude chamber functionality. (T-2). Additional functional checks may be accomplished at the user's discretion. Direct any questions on altitude chambers contract maintenance to 19 AF/DOA.

10.4. Device Transfer and Configuration Control. Maintain device configurations according to 19 AF/DOA guidance. Devices should be transferred upon coordination with 19 AF/DOA in accordance with AFI 16-1007. (T-1).

10.5. Training Device Configuration Change Requests. Complete an AF Form 1067 to request configuration changes to any AP training device and submit to 19 AF/DOA, who will coordinate and forward to the SPO. (T-1).

Chapter 11

TRAINING ADMINISTRATION AND REPORTS

11.1. Documentation of AP Training. Units may use their own computer generated AF Forms prescribed in this AFMAN. Computer-generated AF Forms must mirror the individual AF Form published on the Air Force e-Publishing web site.

11.2. AF Form 1274.

11.2.1. Upon completion of instructor based training, AP personnel will certify and document the training via the AF Form 1274, using the appropriate formal course ID and provide it to the student. The student will present the AF Form 1274 to their local host aviation resource management (HARM) office for update to the database, which will be maintained as physiological training status currency as well as history in the formal training section. A 4M071 or higher will sign any AF Form 1274 documenting completed AP training. (T-2). See **Attachment 2** for guidance on completing the AF Form 1274.

11.2.2. Electronic or hard copies of the AF Form 1274 will be provided to the student and handled in accordance with with **paragraph 11.2** Hard copies may be signed digitally or by hand. Units may email the digitally signed AF Form 1274 to the aircrew members' HARM office if they choose.

11.3. Hazardous Duty Pay.

11.3.1. Per Title 37 USC § 301, Incentive Pay: Hazardous Duty Hazardous duty pay is authorized for personnel performing inside observer duty in a hypobaric chamber. (T-0). Flight surgeons do not qualify for hazardous duty incentive pay.

11.3.2. Units will manage hazardous duty pay in accordance with AFMAN 65-116, Vol 1, Defense Joint Military Pay System Active Component (DJMS-AC) Financial Management Flight (FMF) Procedures and AFMAN 65-116, Vol 2, Defense Joint Military Pay System Active Component (DJMS-AC) Unit Procedures Excluding Financial Management Flights.

11.4. Altitude Chamber Flight Record. Units will maintain record of individual altitude chamber flights. Units may locally develop their form but as a minimum the record will include the following:

11.4.1. Names of AP personnel filling crew positions

11.4.2. Students and their seat and equipment numbers.

11.4.3. Training profile times and altitudes.

11.4.4. All reactions that occur during the training event.

11.4.5. Multiple rapid decompressions may be included on the same sheet of the main altitude chamber flight.

11.5. Reports.

11.5.1. APTU Training Report. APTUs will prepare a quarterly report and send to the MAJCOM AP PM and MFM no later than the 10 days following the end of the quarter i.e., January/April/July/October. (T-2).

11.5.2. MAJCOM AP PM/MFM will consolidate quarterly reports to generate an annual physiological training report and send to 19 AF/DOA by no later than October 10th for the preceding fiscal year. This report is used to assess future Air Force training requirements, procure and assign specialized equipment, provide for the availability of trained officers and enlisted personnel to conduct the program. It also provides case studies for altitude chamber reactions, and monitors the workload at training facilities.

11.5.3. 19 AF/DOA will collate all MAJCOM physiological training reports into a single AF-wide report for AF/A3T to be submitted no later than October 31st for the preceding year's training.

11.5.4. 19 AF/DOA will provide guidance on completing the quarterly report.

Chapter 12

STUDENT REACTOR REPORTING

12.1. Decompression Sickness (DCS). For suspected/confirmed DCS resulting from altitude chamber training the first priority is to manage the student in accordance with the unit's emergency reactor plan. Units will notify 19 AF/DOA within 1 duty day of the incident. 19 AF/DOA will publish guidance on DCS reporting and equipment inspection.

JOSEPH T. GUASTELLA Jr., Lt Gen, USAF
Deputy Chief of Staff, Operations

(PACAF)

Lansing R. Pilch, Maj Gen, USAF
Director of Air and Cyberspace Operations

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

(Added-PACAF) DAFI 33-360, *Publications and Forms Management,* 1 December 2015

Privacy Act (5 U.S.C. § 552a)

37 USC § 301a, *Incentive Pay: aviation career*

37 USC § 301, *Incentive Pay: Hazardous Duty*

PL 93-294, *Aviation Career Incentive Act of 1974*

AIR STD ASM 3003 Ed 1 v3, *Aviation Medicine/Physiological Training of Aircrew*, 23 May 2020

AFPD 10-9, *Lead Command Designation and Responsibilities for Weapon Systems*, 8 March 2007

AFPD 11-2, *Aircrew Operations*, 31 January 2019

AFPD 11-4, *Aviation Service*, 12 April 2019

AFMAN 11-202 Vol 3, *Flight Operations*, 10 June 2020

AFI 11-301, Vol 1. *Aircrew Flight Equipment (AFE) Program*, 10 October 2017

AFI 11-401, *Aviation Management*, 10 December 2010

AFMAN 11-404, *Fighter Aircrew Acceleration Training Program*, 27 November 2019

AFI 11-410, *Personnel Parachute Operations*, 4 August 2008

AFI 16-1007, *Management of Air Force Operational Training Systems*, 1 October 2019

AFI 33-322, *Records Management and Information Governance Program*, 23 March 2020

AFI 33-360, *Publications and Forms Management*, 1 December 2015

AFI 41-117, *Medical Service Officer Education*, 25 March 2015

AFI 48-123, *Medical Examinations and Standards*, 5 November 2013

AFI 91-202, *The US Air Force Mishap Prevention Program*, 12 March 2020

AFI 91-204, *Safety Investigation and Hazard Reporting*, 27 April 2018

AFMAN 11-202, Volume 1, *Aircrew Training*, 27 September 2019

AFMAN 65-116, Vol 1, *Defense Joint Military Pay System Active Component (DJMS-AC) Financial Management Flight (FMF) Procedures*, 23 October 2019

AFMAN 65-116, Vol 2, *Defense Joint Military Pay System Active Component (DJMS-AC) Unit Procedures Excluding Financial Management Flights*, 23 December 2019

DoDI 5400.11, *DoD Privacy and Civil Liberties Programs*, 29 January 2019

DoDI 7730.67, *Aviation Incentive Pays and Bonus Program*, 20 October 2016

INFO PUB ASM 3009 Ed 1 v3, *Spatial Disorientation*, 23 May 2020

NATO STANAG 3114, *Aeromedical Training of Flight Personnel*, 5 February 2018

NATO STANAG 3475, *Temporary Flying Restrictions Due to Exogenous Factors Affecting Aircrew Efficiency*, 19 July 2018

00-20-1, *Aerospace Equipment Maintenance Inspection, Documentation, Policies, and Procedures*, 15 June 2013

43D8-3-2-81, *Hypobaric Training Assembly*, 1 November 2015

Prescribed Forms

(**Added**-**PACAF**) None

Adopted Forms

(**Added**-**PACAF**) AETC Form 620, *Academic Instructor Monitoring Checklist*

AF Form 1274, Physiological Training

(**Added**-**PACAF**) AF Form 1522, *ARMS Additional Training Accomplishment Report*

AF Form 847, Recommendation for Change of Publication

DD Form 2992, Medical Recommendation for Flying or Special Operational Duty

AF Form 1067, Modification Proposal

AF Form 4026, Aircrew Training Devices Utilization

AFTO Form 95, Significant Historical Data

AFTO Form 244, Industrial/Support Equipment Record

AFTO Form 334, Helmet and Oxygen Mask/Connector Inspection Data

Abbreviations and Acronyms

AED—Automated External Defibrillator

AETC—Air Education and Training Command

AF—Air Force

AFE—Aircrew Flight Equipment

AFI—Air Force Instruction

AFLCMC—Air Force Life Cycle Management Center

AFMAN—Air Force Manual

AFMS—Air Force Medical Services

AFOSH—Air Force Occupational Safety and Health

AFPD—Air Force Policy Directive

AFPC—Air Force Personnel Center

AFR—Air Force Reserve

AFRIMS—Air Force Records Information Management System

(Added-PACAF) AFSAS—Air Force Safety Automated System

AFSAT—Air Force Security Assistance Training

AFSC—Air Force Specialty Code

AFSEC—Air Force Safety Center

AFSPC—Air Force Space Command

(Added-PACAF) AGSM—Anti-G Straining Maneuver

ANG—Air National Guard

AP—Aerospace Physiology

APO—Aerospace Physiology Officer

APTF—Aerospace Physiology Training Flight

APTT—Aerospace Physiology Training Team

APTU—Aerospace Physiology Training Unit

ARC—Air Reserve Component

ARMS—Aviation Resource Management System

(Added-PACAF) ASAP—Aviation Safety Action Program

BSC—Biomedical Science Corps

CAP—Civil Air Patrol

C4ISR—Command Control Communications Computers Intelligence Surveillance and Reconnaissance

CC—Commander

CCR—Course Change Request

CFETP—Career Field Education and Training Plan

COR—Contracting Officer's Representative

CPR—Cardiopulmonary Resuscitation

CUI—Controlled Unclassified Information

DCS—Decompression Sickness

DNIF—Duty Not Including Flying

DoD—Department of Defense

DRU—Direct Reporting Unit

DV—Distinguished Visitor

FAA—Federal Aviation Administration

(Added-PACAF) GPBAT—Ground Pressure Breathing for Altitude Trainer

HAF—Headquarters Air Force

HARM—Host Aviation Resource Management

HFT—Hypoxia Familiarization Trainer

HQ—Headquarters

(Added-PACAF) HUD—Heads-Up Display

IAW—In Accordance With

I/O—Instructor/Operator

IT—Information Technology

ITO—International Travel Order

LDT—Lateral Drift Trainer

(Added-PACAF) LOSA—Line Operations Safety Audit

(Added-PACAF) MAAF—Mishap Analysis and Animation Facility

MAJCOM—Major Command

MDS—Mission Design Series

MFF—Military Free Fall

MFM—MAJCOM Function Manager

(Added-PACAF) MFOQA—Military Flight Operations Quality Assurance

NATO—North Atlantic Treaty Organization

N/A—Not Applicable

(Added-PACAF) NVD—Night Vision Device

OG—Operations Group

OPR—Office of Primary Responsibility

OSS—Operational Support Squadron

(Added-PACAF) PBA—Pressure Breathing for Altitude

PPE—Personal Protective Equipment

(Added-PACAF) RAP—Ready Aircrew Program

ROBD—Reduced Oxygen Breathing Device

ROTC—Reserve Officer Training Corps

RPA—Remotely Piloted Aircraft

(Added-PACAF) RTM—Ready Aircrew Program (RAP) Tasking Message

SPO—System Program Office

STANAG—Standardization Agreement

Stan/Eval—Standardization/Evaluation

UFT—Undergraduate Flight Training

USAF—United States Air Force

USAFA—United States Air Force Academy

Terms

Aerospace Physiology (AP) Personnel—The total complement of officer and enlisted members holding the AFSC of 43AX or 4M0X1 responsible for providing aerospace physiology academic training and the safe operation of physiology training devices.

Aircrew—Personnel responsible for the safe ground and flight operation of the aircraft and onboard systems, or for airborne duties essential to the accomplishment of the aircraft's mission. It includes the total complement of rated pilots (manned and unmanned), navigators, combat systems operators, air battle managers, and flight surgeons, as well as career enlisted aviators, and nonrated aircrew. Members in initial formal training for immediate assignment to an authorized operational flying operation are also included. See AFPD 11-4.

Evaluation—1) Procedure to determine the effectiveness of the performance of an instructional product or process in order to ascertain specific causes for the effectiveness or lack of it, and to make decisions appropriate to the extent of the effectiveness. Evaluation of formal courses includes field evaluation and internal evaluation. 2) The review and analysis of qualitative or quantitative data obtained from design review, hardware inspection, testing or operational use of equipment.

Lead Command—The Air Force MAJCOM or agency possessing an MDS that is designated by AFPD 10-9 as responsible for the coordination of MDS-Specific activities.

Office of Primary Responsibility (OPR)—Any headquarters, agency, or activity having the primary functional interest in, and responsibility for, a specific action, project, plan, program or problem.

Read File—A collection of publications and material determined by the MAJCOM and unit as necessary for day-to-day operations.

Weapon System—A combination of one or more weapons with all related equipment, materials, services, personnel, and means of delivery and deployment (if applicable) required for self-sufficiency.

Attachment 2

APTU FORMS AND COMPLETION STANDARDS

A2.1. General. This attachment references guidance for the completion of forms required to document AP training. Request changes to required forms to 19 AF/DOA using the AF Form 847.

A2.2. Forms. Maintain forms in accordance with AFMAN 33-322, Records Management and Information Governance Program, guidance using Air Force Records Information Management System (AFRIMS) database for disposition schedule. For convenience, forms disposition per AFRIMS is incorporated in this manual. Changes to disposition via AFRIMS supersede any disposition schedule for AP records described in this manual.

A2.3. AF Form 1274. Use AF Form 1274 to document training directed by this document or AFMAN 11-404. A complete listing of Formal Course IDs can be found in **Table A2.1** Do not use AF Form 1274 to document Cadet training, training required by other AFIs, or local requirements. APTUs are not required to maintain copies of completed AF Form 1274s but are required to maintain a list of names and pertinent information on students who received training. This list will be maintained using the 19 AF/DOA provided guidance. See **Table A2.1** for AP Training Formal Course ID codes.

Table A2.1. AP Training Formal Course ID codes.

Formal Course ID	Training Course
S-O-B/A-APH-I	Initial Physiological Hypoxia Training
AP RPA-I	Initial RPA AP Training (no hypoxia exposure)
AP RPA-R	Refresher RPA AP Training (no hypoxia exposure)
S-O-B/A-APH-T	Transition Physiological Hypoxia Training
S-O-B/A-APH-R	Refresher Physiological Hypoxia Training – all Tracks

Figure A2.1. Completed 1274.



PHYSIOLOGICAL TRAINING

This is to certify that the following person has met the requirements for the USAF Physiological Training Program.

Last Name, First Name, MI.
Jones, Indiana, E.J.

TRAINING COMPLETED
Track J

GRADE	FORMAL COURSE ID
O-3	S-O-B/A-APH-R

ISSUE DATE	EXPIRATION DATE
20180620	20230630

PHYSIOLOGICAL TRAINING UNIT
1 SOSS/OSM, Hurlburt Field, FL

SIGNATURE OF 43A/4M071
Click to sign

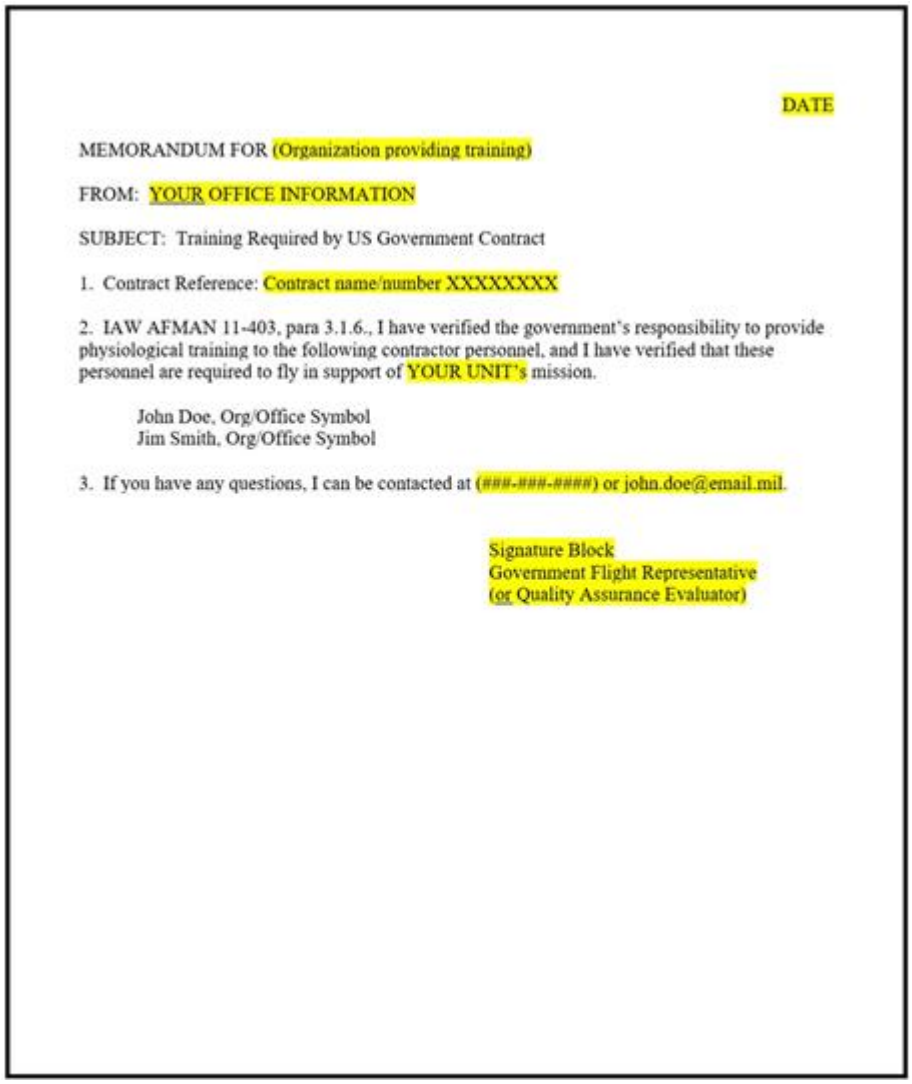
Signature Block: First MI. Last Name, USAF, BSC
ISMAEL PAEZ JR., SMSgt, USAF

AF FORM 1274, 20190620
Previous Edition Will Not Be Used.

Attachment 3

SAMPLE CONTRACTOR TRAINING MEMORANDUM

Figure A3.1. Contractor Training Memorandum.



DATE

MEMORANDUM FOR (Organization providing training)

FROM: YOUR OFFICE INFORMATION

SUBJECT: Training Required by US Government Contract

1. Contract Reference: Contract name/number XXXXXXXX

2. IAW AFMAN 11-403, para 3.1.6., I have verified the government's responsibility to provide physiological training to the following contractor personnel, and I have verified that these personnel are required to fly in support of YOUR UNIT's mission.

John Doe, Org/Office Symbol
Jim Smith, Org/Office Symbol

3. If you have any questions, I can be contacted at (###-###-####) or john.doe@email.mil.

Signature Block
Government Flight Representative
(or Quality Assurance Evaluator)